LAMOTRIGINE

Slow titration — rash/SJS

Safer in pregnancy

CHEMICAL NAME 6-(2,3-dichlorophenyl)-1,2,4-triazine-3,5-diamine
CLASS Phenyltriazine antiepileptic and mood stabiliser
TRADE NAMES Lamictal · Lamotrix · Lamogine

THERAPEUTIC USES / INDICATIONS

- Focal and generalised seizures (incl. Lennox–Gastaut); preferred in women of childbearing age.
- Bipolar disorder; prevention of depressive episodes.

ADMINISTRATION

- **PO** once or twice daily; dispersible tablets can be chewed or dispersed in water.
- Strict slow titration schedule (depends on co-medication).

DOSAGES

- **Monotherapy:** 25 mg daily for 2 weeks, 50 mg daily for 2 weeks, then increase by 50–100 mg every 1–2 weeks; usual 100–200 mg/day.
- **With valproate:** 25 mg alternate days for 2 weeks, then 25 mg daily; usual 100–200 mg/day.
- **With enzyme inducers (not valproate):** 50 mg daily, usual 200–400 mg/day.
- If stopped >5 half-lives, restart titration from the beginning.

PHARMACOKINETICS

- **Peak:** 1–3 h. Bioavailability ~98%.
- Glucuronidated in liver (↑ by oestrogens, inducers; ↓ by valproate).
- Excreted in urine. **t½:** 25–35 h (70 h with valproate; 14 h with inducers).

PHARMACODYNAMICS

- Blocks **voltage-gated sodium channels** → ↓ glutamate release.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache, dizziness, diplopia, ataxia.
- Nausea, insomnia.
- Benign rash (common).

ADVERSE EFFECTS (SERIOUS)

- Serious rash: SJS/TEN (usually first 8 weeks; ↑ with fast titration, valproate, children).
- HLH (haemophagocytic lymphohistiocytosis).
- Aseptic meningitis; blood dyscrasias.
- Cardiac conduction effects (Brugada).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity; previous lamotrigine rash.
- **Caution:** valproate co-therapy, cardiac conduction disease.

KEY INTERACTIONS

- **Valproate:** doubles lamotrigine levels.
- **Combined hormonal contraceptives, pregnancy:** ↓↓ lamotrigine levels (dose adjust; levels rise after stopping/delivery).
- **Carbamazepine, phenytoin, rifampicin:** ↓ levels.

NURSING MANAGEMENT

- Follow titration exactly; check co-medication.
- Any rash: stop and seek urgent review.
- Monitor seizure control and mood.
- Pregnancy: monitor levels and adjust.

PATIENT EDUCATION

- Increase dose only as instructed.
- Seek urgent advice for any rash, fever, mouth ulcers, swollen glands.
- Tell your doctor if you start/stop the pill or become pregnant.
- Do not stop suddenly.

KEY POINTS

- Na^+-channel blocker; good in women.
- Halve dose with valproate.
- Slow titration — SJS risk.
- Pill and pregnancy lower levels.

SOURCES CHECKED [729] Lamotrigine 100 mg tablets (UK SmPC) [730] Lamotrigine, oral (US label) [731] Lamotrigine, oral (US label, Amneal Pharmaceuticals of New York LLC)
Full references p. 432–457